EXPERIMENTAL INVESTIGATION OF THE CHEMISTRY OF EXCITED STATES OF RARE GASES

Quarterly Progress Report
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D. W. Setser

Chemistry Department
Kansas State University
Manhattan, KS 66506

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Third Quarterly Progress Report

The personnel working on the E(11-1)-2807 ERDA contract are Mr. J. E. Velazco (graduate student) and Dr. J. Rodriguez (postdoctoral fellow). Mr. Velazco has been involved with measuring rate constants for formation of xenon halide excimers. Mr. Rodriguez has been studying the laser induced fluorescence of Ar*(4p) and Ar*(5p) states.

The measurement of the total quenching rate constants and the branching ratios for XeF^* or KrF^* formation from reactions of $Kr(^3P_2)$ and $Xe(^3P_2)$ metastable rare gas atoms with fluorine-containing molecules is completed, and the manuscript describing that work has been accepted by J. Chem. Phys. Mr. Velazco has completed a less extensive study of $Xe(^3P_2)$ with other halogen donors (RI, RCl, RBr). The trends are very similar, i.e. only the diatomic halogens have unit efficiency for excimer formation. All other Cl, Br and I donors have significantly reduced branching ratios.

Mr. Velazco also has studied the reactions of $Ar(^3P_2)$, $Kr(^3P_2)$ and $Xe(^3P_2)$ with XeF_2 and a manuscript is being written for Chem. Phys. Letts., which will describe the results. In each reaction XeF^* (both $^2\Sigma^+$ and $^2\Pi_{1/2}$) is the principal exit channel, although ArF^* and KrF^* also are formed by the $Ar(^3P_{0,2})$ and $Kr(^3P_2)$ reactions. For $Xe(^3P_2)$ and $Kr(^3P_2)$ the branching ratios for XeF* formation must be nearly unity These reactions provide an excellent source for obtaining spectra of the $XeF^*(^2\Sigma^+-^2\Sigma^+)$ and $XeF^*(^2\Pi_{1/2}-^2\Sigma^+)$ bands. Photographs have been obtained and are being shared with Dr. Tellinghuisen, who is trying to finalize the description of these XeF potential curves. Mr. Velazco also is writing his Dissertation.

Dr. Rodriguez has used a tunable dye laser to pump argon atoms from the metastable $(4s,^3P_0)$ or $4s,^3P_2)$ states up to the $Ar^*(4p)$ or $Ar^*(5p)$ levels. The subsequent fluorescence from these levels is used to deduce decay constants for these $Ar^*(4p)$ and $Ar^*(5p)$ levels in an argon bath gas. A summary of this work constitutes the main body of this report.

Lifetimes and Two-body Quenching Rate Constants of Ar*(4p) and Ar*(5p) States

This work has not been described in previous quarterly reports. Therefore a fairly detailed discussion follows.

Introduction

Collisional quenching rate constants and radiative lifetimes of excited states of argon and other rare gases are important for understanding the energy transfer steps pertinent to the operation of electrically pumped lasers using argon as the principal gas. The direct excitation of individual excited states of argon with a pulsed light source followed by observation of the fluorescence is a convenient method for the determination of lifetimes and deactivation rate constants. A tunable dye laser pumped by a nitrogen laser provides this high intensity source. This source in combination with a hollow-cathode discharge attached to a flowing-afterglow apparatus permits the Ar*(5p) and Ar*(4p) states to be produced by tuning the dye laser to the absorption lines of $Ar(^3P_2)$ or $Ar(^3P_0)$, see Figure 1. The kinetics of the Ar*(4p) and Ar*(5p) levels can be monitored by observing the laser induced fluorescence. The notation for the excited rare gas atomic states is somewhat confusing. The four states arising from the $Ar(3p^5 4s)$ configuration are usually named according to L-S coupling as $^3P_2$, $^3P_1$, $^3P_0$, $^1P_1$ in order of increasing energy. The more proper J_c-1 coupling notation is $4s[3/2]_2$, $4s[3/2]_1$, $4s'[1/2]_0$ and $4s'[1/2]_1$; whereas, the Paschen numbering is $1s_5$, $1s_4$, $1s_3$ and $1s_2$. The higher states are normally identified by the J_c-1 notation or the Paschen numbering. The present report describes the experimental technique, as well as some results obtained by the excitation of the Ar*(5p) and Ar*(4p) states. No interpretation of the data will be attempted.

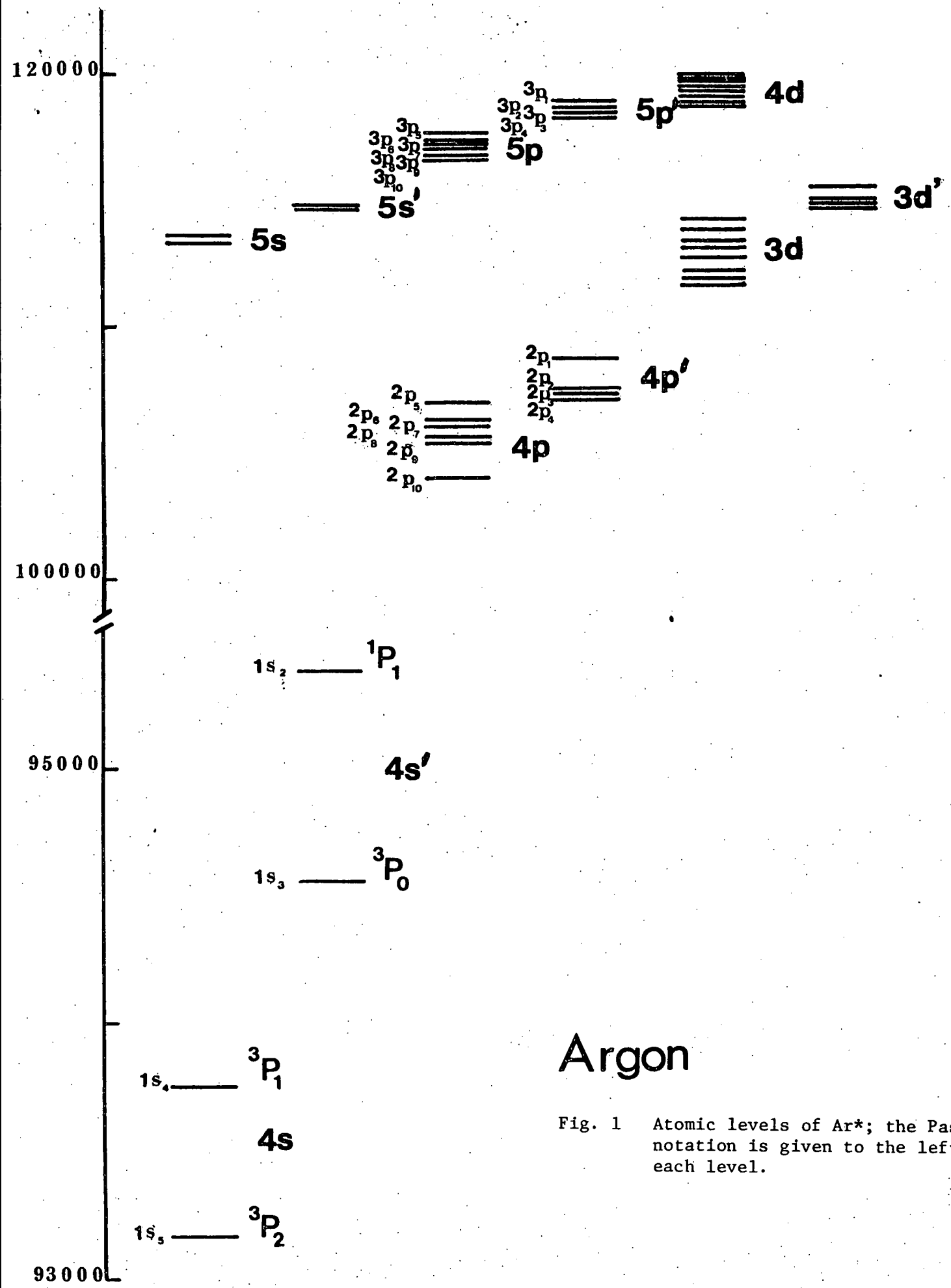


Fig. 1 Atomic levels of Ar*; the Paschen notation is given to the left of each level.

Experimental

The apparatus consisted of our usual hollow-cathode discharge flow system for generating $Ar(^3P_2)$ and $Ar(^3P_0)$ atoms at a concentration of $\sim 5 \times 10^{10}$ atoms cm^{-3}. The metastable atoms, highly diluted in Ar, flowed into a laser induced fluroescence cell that had two baffle arms for the entrance and exit of the laser light and two viewing windows. One window was backed with a mirror and the other with a suitable filter to isolate the laser induced fluorescence. A RCA 7265 photomultiplier tube viewed the fluorescence zone through various filters chosen to best isolate the laser induced fluorescence. The signal of the photomultiplier tube was averaged by a PAR 162 boxcar integrator with two model 164 channel modules. One channel (B) observed only the peak of the fluorescence signal; while the other (A) scanned and averaged the signal as a function of time. In order to compensate for fluctuations in the incident laser intensity, the output of channel A over channel B was recorded. The laser system was a homemade Hansch type dye laser pumped by an AVCO 250 kwatt nitrogen laser. The laser was operated at 30 to 50 pps; the bandwidth was less than 0.1 Å and the pulse duration was ~ 5 nsec. The trigger circuit from the power supply module was used to trigger the boxcar and the oscilloscope. The desired wavelength from the dye laser was selected by adjustment of the grating forming one end of the dye laser cavity and monitored with a 1/2 meter Jarrel Ash monochromator.

For the $Ar^*(5p)$ states, the scintillators 0.001 M 1,4 bis[2-(5-phenyl-oxazolyl)benzene in p-dioxane and 7×10^{-4} M 1,4 bis[2-(4-methyl-5-phenyl-oxazolyl) benzene in p-dioxane were found to be appropriate dyes for the spectral range, 407.0 - 451.5 nm and 418.0 - 465.0 nm, respectively. These dyes are commonly called POPOP and dimethyl-POPOP. A mixture of rhodamine-6G and cresyl violet perchlorate in ethanol was used as the dye for the study of the $Ar^*(4p)$ states. Only the three highest energy states of the 4p manifold could be studied because of the long wavelength limitation of the dye; the wavelength range was from 714.7 to 696.5 nm.

Results and Analysis

A. Argon 4p states.

The experiments with the Ar*(4p) levels were very straightforward because only the fluorescence from the level that was pumped was monitored. The transitions from the $2p_2$, $2p_3$, and $2p_4$ levels to $4s'[1/2]_1$ following laser excitation was isolated from other transitions with an interference filter. The fluorescence decay curves were simple exponentials and were analyzed to yield pseudo first order rate constants. These decay curves were obtained for a range of pressure, and a plot of the observed decay constants vs argon pressure are shown in Figures 2 for the $4p'[3/2]_1$ level. Such plots were fitted to a least squares line, and the zero pressure intercepts and slopes are tabulated in Table 1. The standard error limits are ±15%. Comparison of the lifetimes of the three Ar*(4p) states, which is the inverse of the zero pressure intercepts, to other measured and calculated lifetimes are given below:

Lifetimes of some 4p Levels

Level	λnm	Present Work (n sec)	Previous Work (n sec)			
			(a)	(b)	(c)	(d)
$2p_2$	696.5	33.5 ± 4	25 ± 2	25.7	24 ± 3	---
$2p_3$	706.7	36.3 ± 4	26 ± 1	27.6	25 ± 3	---
$2p_4$	714.7	45.2 ± 5	31 ± 2	29.0	27 ± 3	37 ± 1

(a) J. Z. Kloze, J. Opt. Soc. Am. 58, 1200 (1968).

(b) R. A. Lilly, J. Opt. Soc. Am. 66, 245 (1976). (calculated values)

(c) J. B. Shumaker, Jr., and C. H. Popenone, J. Opt. Soc. Am. 57, 8 (1967).

(d) J. P. Grandin, D. Lecler, J. Margerie, Acad. Sci. (Paris) C. R. (A&B) 272, 929 (1971). In addition to this absolute measurement there is fragmentatory evidence, usually based upon oscillator strengths of a particular line, that also favor longer lifetimes than given by a,b, and c.

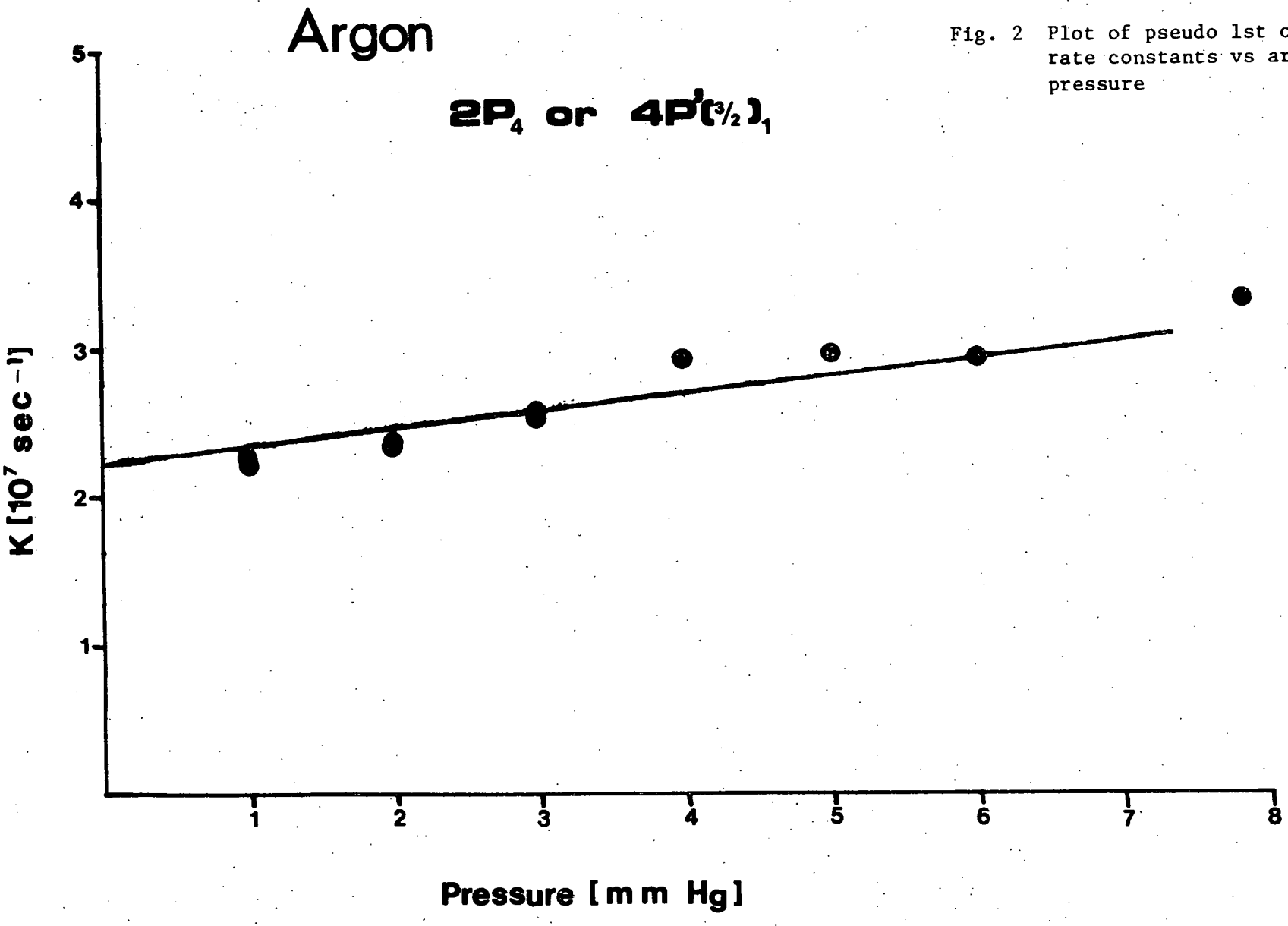


Fig. 2 Plot of pseudo 1st order rate constants vs argon pressure

Table 1 Argon 4p States

Level	laser wavelength[a] nm	k_o or τ^{-1} (sec^{-1})	Rate Constant (cm^3 $atom^{-1}$ sec^{-1})
$2p_2$ or $4p'[1/2]_1$	696.54	2.92×10^7	5.21×10^{-11}
$2p_3$ or $4p'[3/2]_2$	706.72	2.74×10^7	6.62×10^{-11}
$2p_4$ or $4p'[3/2]_1$	714.70	2.22×10^7	3.69×10^{-11}

a. In each case the upper level was produced by pumping from the lower metastable Ar(4s) level, see Fig 1.

Table 2 Argon 5p States

Level	laser wavelength[a] nm	k_o or τ^{-1} (sec^{-1})	Rate Constant (cm^3 $atom^{-1}$ sec^{-1})
$3p_6$ or $5p[3/2]_2$	415.86	0.78×10^7	5.15×10^{-11}
$3p_7$ or $5p[3/2]_1$	416.42	0.83×10^7	5.89×10^{-11}
$3p_8$ or $5p[5/2]_2$	419.07	0.81×10^7	5.15×10^{-11}
$3p_9$ or $5p[5/2]_3$	420.07	0.85×10^7	5.62×10^{-11}
$3p_{10}$ or $5p[1/2]_1$	425.12	0.75×10^7	5.75×10^{-11}
$3p_2$ or $5p'[1/2]_1$	418.2	0.16×10^7	3.74×10^{-11}

The $3p_6$ - p_{10} levels were produced by pumping from the lower metastable Ar($^3P_2$) level, see Fig 1. The p_2 level was produced by pumping from the upper metastable, Ar($^3P_0$), level.

The difference between our values and the others appear to be outside the random error limits and suggest a systematic difference. As noted by Wiese Smith and Miles in their NBS tabulation, the experimental lifetime data tend to fall into two groups. Our data appear to support the longer lifetime measurements. These Ar*(4p) studies were our first experiments and several improvements in apparatus and technique subsequently have been made. Therefore, we intend to reinvestigated these levels over a wider pressure range. There is no apparent reason why the uncertainty cannot be reduced to $\pm$5%. If there is systematic error, it should affect the intercepts more than the rate constants. Since a single Ar*(4p) level was monitored, the two body quenching rate constants are for all collision processes that remove atoms from the particular level being observed. The main processes presumably are collisional transfer to the lower energy levels of the Ar*(4p) manifold. Experiments to identify product states by observation of emission from other Ar*(4p) levels were unsuccessful. One of the difficulties was the background emission generated by the hollow-cathode discharge, which contributes to the noise level observed by the boxcar. With our recent improvements, it may be possible to identify the individual Ar(4p) product states. This possibility will be checked in our reinvestigation of these levels.

B. Argon 5p states

Pumping the Ar*(5p) levels with the laser is easier than the Ar*(4p) levels because of the superior performance of the dye laser in this wavelength range. However, despite considerable effort we were unable to observe the laser induced fluorescence from the excited Ar*(5p) level back to the Ar($^3P_1$ or $^1P_1$) levels. In fact, the 5p-4s fluorescence was not even observable with band-pass filters that transmitted <u>all</u> of the emission from the 5p-4s transitions. Therefore, an indirect observation scheme was adopted. This is demonstrated in Figure 3. The observed fluorescence

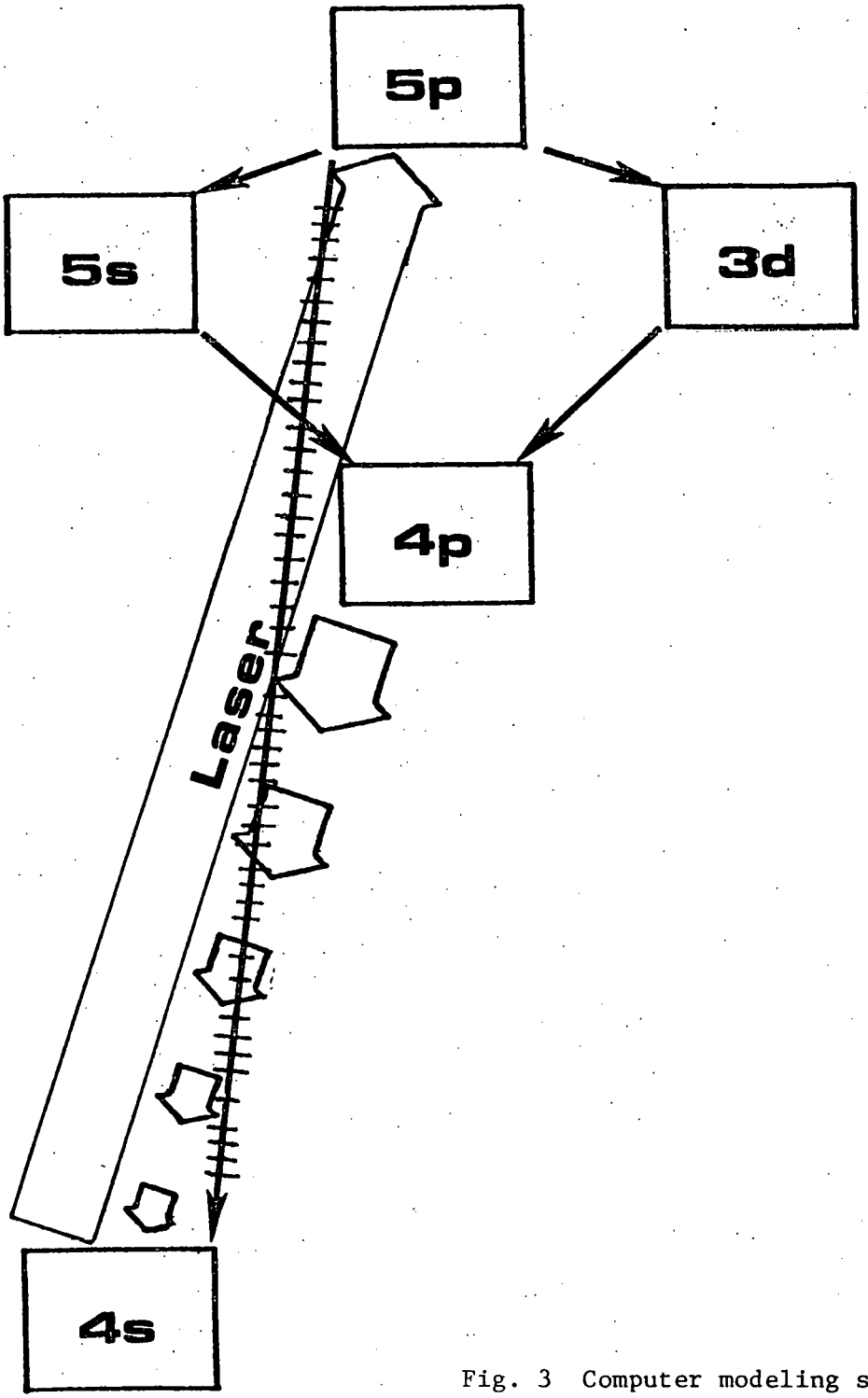


Fig. 3 Computer modeling scheme used to analyze the 5p states. The 5p-4s transitions could not be observed.

was transitions from the 4p levels to the 4s levels at long wavelengths. This long wavelength emission intensity was monitored with a wide band-pass filter, which permitted the photomultiplier to view all emission originating from the 4p levels.

Figure 4 shows a typical decay plot obtained by monitoring the emission intensity following excitation of $Ar^*(5p[3/2]_2)$. Detailed computer modeling of the process indicated in Figure 3 was done by considering the equations below.

$$Ar^*(5p) \xrightarrow{\tau^{-1}} Ar^*(3d \text{ and } 4s) + h\nu$$

$$\xrightarrow{k_Q} Ar^*(3d \text{ and } 5s)$$

$$Ar^*(3d \text{ and } 5s) \xrightarrow{\tau^{-1}} Ar^*(4p) + h\nu$$

$$Ar^*(4p) \xrightarrow{\tau^{-1}} Ar^*(4s) + h\nu$$

Collisional quenching of the $Ar^*(3d + 5s)$ and $Ar^*(4p)$ can be included in the scheme; however, this does not effect the interpretation of the long-term decay component of Figure 4. Computer modeling of the reaction scheme indicated that the decay curve was controlled by the kinetics of the 5p states, because the radiative lifetimes of the $Ar^*(3d \text{ and } 5s)$ and $Ar^*(4p)$ are shorter than for $Ar^*(5p)$. The long-term decay was simply exponential and the first order rate constants, derived from just the long-term portion of the decay curves, are shown in Figure 5 for $Ar^*(5p[5/2]_2)$. Extracting the rate constant from the long-term part of the decay curves at higher pressure is not straightforward because of the increase in magnitude of the rate constant and because of poor signal to noise ratio at the higher pressures. The failure of the simple analysis is indicated by the low values of the first order rate constants at higher pressures shown in Figure 6. Table 2 (see page 7) summarizes the collisional quenching rate constants ($5p \rightarrow (5s + 3d)$ levels) and the radiative lifetimes of the 5p states. These results are based upon fitting the data at pressure below 8 torr to least squares straight lines.

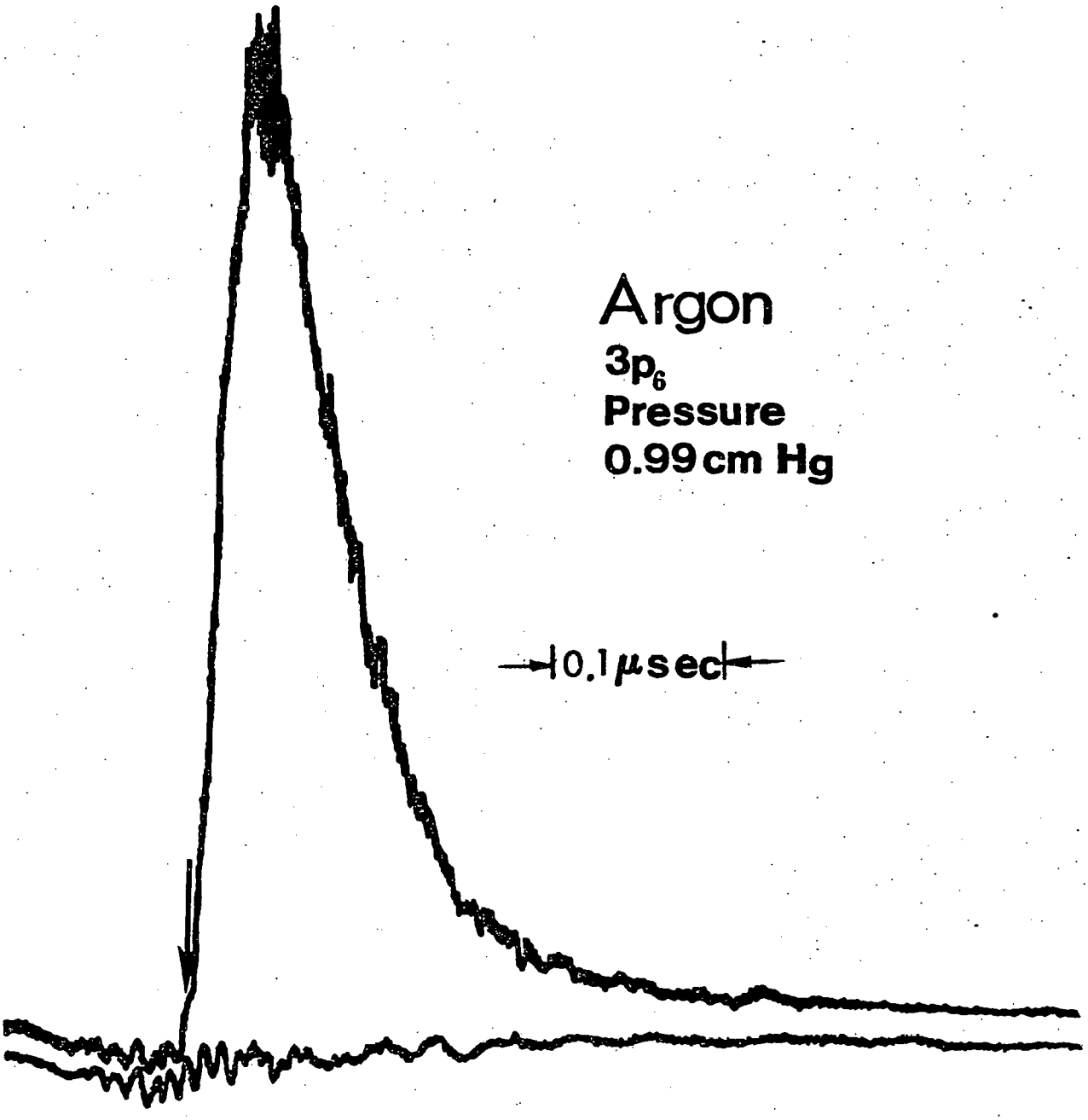


Fig. 4 The arrow denotes the location of the laser pulse, which is zero time. The delay between the laser pulse and the maximum in the fluorescence signal was ∿50n sec.

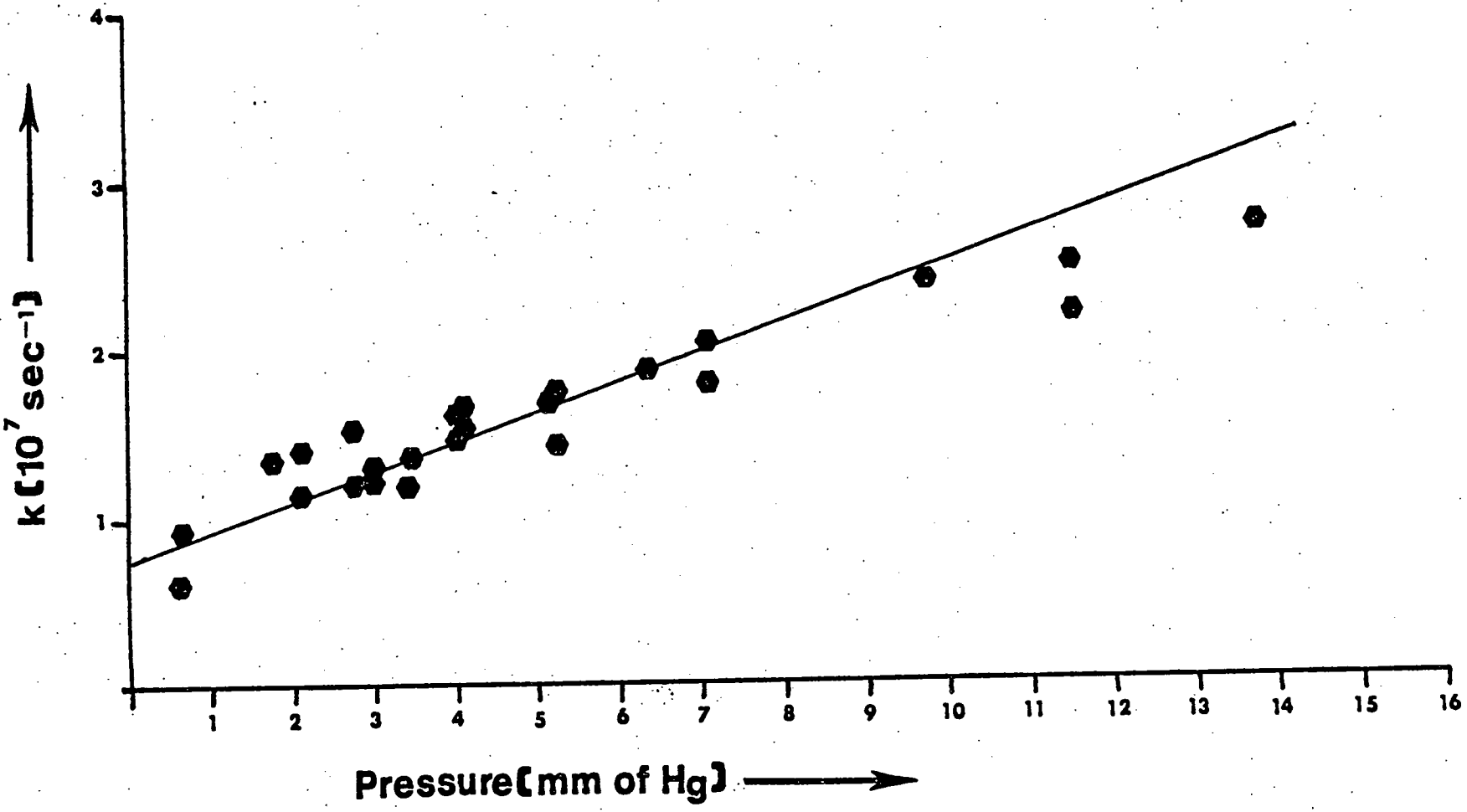


Fig. 5 Plot of Pseudo First Order Rate Constants vs Argon Pressure

The common value of the two-body rate constants implies that collisional relaxation within the $Ar^*(5p)$ levels occurs prior to transfer to the 5s and 3d levels. Thus, the measured values are the rate constant for transfer from the manifold of 5p levels to the 3d and 5s manifolds. By fitting the entire decay plot, rather than using just the long-term portion, it should be possible to extract a more reliable two-body quenching rate constant. We fully expect this more complete data treatment to restore the points in the 8-16 torr range to the linear plot. Based upon the data in hand, we see no evidence for three-body quenching that would effectively compete with the two-body transfer from the 5p manifold to the 5s and 3d manifolds. An attempt will be made to model the rise time of the decay plots, see Fig. 4, to extract information about the processes converting the 3d + 5s levels to the 4p levels. The <u>failure</u> to observe fluorescence from the 5p levels back to the 4s levels is particularily <u>hard to understand</u>. According to the calculated radiative branching ratios for the transitions from the 5p levels to the 5s, 3d and 4s levels, the emission from $Ar^*(5p)$—$Ar^*(4s)$ should have been observable. The intercept values from the $Ar^*(5p)$ experiments are very close to the calculated radiative lifetimes. The tentative explanation, very fast collisional transfer within the $Ar^*(5p)$ manifold,is not totally acceptable. As an experimentalist, another tempting explanation is that the calculated branching ratios may be somewhat in error. Further work remains to be done in order to understand these apparently "simple" experiments.

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